

Positive Parallel Cuts Give a Head Upper Bound

Setup

Work on the Boolean cube $\{0, 1\}^n$. A Boolean function is a map

$$f : \{0, 1\}^n \rightarrow \{0, 1\}.$$

The quantity $H^*(f)$ is the minimum number of attention heads needed by a single-layer attention-only model, with a final linear threshold readout, to compute f exactly on the cube.

For the present note, we only need the following scalar normal form for one head. Choose parameters

$$\gamma > 0, \quad \rho_i > 0, \quad \alpha > 0, \quad \alpha \neq 1,$$

and choose arbitrary real parameters

$$\eta, \quad m_1, \dots, m_n, \quad \delta.$$

After composing the head output with the final linear readout, a single head can contribute the scalar atom

$$\phi(x) = \frac{\eta + \sum_{i=1}^n \rho_i \alpha^{x_i} (m_i + \delta x_i)}{\gamma + \sum_{i=1}^n \rho_i \alpha^{x_i}},$$

where $x \in \{0, 1\}^n$.

Let H be a nonnegative integer. To prove $H^*(f) \leq H$, it is enough to choose H atoms

$$\phi_1, \dots, \phi_H$$

and a constant $c \in \mathbb{R}$ such that the real score

$$S(x) = c + \sum_{h=1}^H \phi_h(x)$$

satisfies

$$f(x) = 1 \iff S(x) > 0 \quad \text{for every } x \in \{0, 1\}^n.$$

Positive parallel cuts

A positive parallel statistic is an affine-linear statistic with strictly positive variable coefficients:

$$t(x) = \sum_{i=1}^n \lambda_i x_i, \quad \lambda_i > 0.$$

Its level sets

$$t(x) = a$$

for real values a are parallel affine hyperplanes in $\mathbb{R}^n$. The positivity assumption means that all these hyperplanes have normal vector in the positive orthant.

Suppose there is a function

$$F : \text{Im}(t) \rightarrow \{0, 1\}$$

such that f factors through t :

$$f(x) = F(t(x)).$$

Write the distinct values of t on the cube in increasing order. Thus, for some integer $M \geq 1$, there are real numbers $\tau_0, \dots, \tau_{M-1}$ with

$$\text{Im}(t) = \{\tau_0 < \tau_1 < \dots < \tau_{M-1}\}.$$

For each $m \in \{0, \dots, M-1\}$, define the sign

$$\sigma_m = \begin{cases} +1, & F(\tau_m) = 1, \\ -1, & F(\tau_m) = 0. \end{cases}$$

The number of label changes along this one-dimensional profile is

$$C_t(F) := |\{m \in \{1, \dots, M-1\} : \sigma_{m-1} \neq \sigma_m\}|.$$

Let $\mathcal{A}_+(f)$ be the set of pairs (t, F) satisfying all of the conditions just stated:

$$\mathcal{A}_+(f) := \left\{ (t, F) : t(x) = \sum_{i=1}^n \lambda_i x_i, \lambda_i > 0, F : \text{Im}(t) \rightarrow \{0, 1\}, f = F \circ t \right\}.$$

For every pair $(t, F) \in \mathcal{A}_+(f)$, the number $C_t(F)$ is defined as above. Define the positive-parallel complexity by minimizing this count:

$$P_{\parallel}^+(f) := \min_{(t, F) \in \mathcal{A}_+(f)} C_t(F).$$

This minimum is finite. For example,

$$t(x) = \sum_{i=1}^n 2^{i-1} x_i$$

is injective on $\{0, 1\}^n$, so every f factors through this positive statistic.

Theorem. *For every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,*

$$H^*(f) \leq P_{\parallel}^+(f).$$

Proof

Fix a pair $(t, F) \in \mathcal{A}_+(f)$. By definition of $\mathcal{A}_+(f)$, there are positive coefficients $\lambda_1, \dots, \lambda_n$ such that

$$t(x) = \sum_{i=1}^n \lambda_i x_i, \quad \lambda_i > 0,$$

and $f(x) = F(t(x))$ for every $x \in \{0, 1\}^n$.

Let M be the number of distinct values of t on the cube. Write these values as

$$\text{Im}(t) = \{\tau_0 < \tau_1 < \dots < \tau_{M-1}\}.$$

For each $m \in \{0, \dots, M-1\}$, let

$$\sigma_m = \begin{cases} +1, & F(\tau_m) = 1, \\ -1, & F(\tau_m) = 0. \end{cases}$$

Let

$$C := C_t(F).$$

We prove

$$H^*(f) \leq C.$$

The theorem follows by minimizing over all positive-parallel representations.

If $C = 0$, then F is constant on $\text{Im}(t)$, so f is constant on the cube. Thus $H^*(f) = 0$. From now on assume $C \geq 1$.

Step 1. A univariate sign polynomial

Let

$$J := \{m \in \{1, \dots, M-1\} : \sigma_{m-1} \neq \sigma_m\}.$$

Thus $|J| = C$. For every $m \in J$, choose a cutpoint

$$a_m \in (\tau_{m-1}, \tau_m).$$

Define

$$p(z) := \sigma_0 \prod_{m \in J} (a_m - z).$$

Then p changes sign exactly once between τ_{m-1} and τ_m for each $m \in J$, and it changes sign nowhere else between consecutive image values. Hence

$$\text{sgn } p(\tau_m) = \sigma_m \quad \text{for every } m = 0, \dots, M-1.$$

Equivalently,

$$F(\tau_m) = 1 \iff p(\tau_m) > 0.$$

Step 2. Put the sign polynomial over a positive denominator

Choose distinct positive shifts

$$r_1, \dots, r_C > 0$$

and set

$$B(z) := \prod_{j=1}^C (z + r_j).$$

Since every τ_m is nonnegative,

$$B(\tau_m) > 0 \quad \text{for every } m.$$

Therefore the rational function

$$R(z) := \frac{p(z)}{B(z)}$$

has the same signs as p on the finite set $\text{Im}(t)$:

$$\text{sgn } R(\tau_m) = \sigma_m.$$

Because $\deg p = \deg B = C$ and the roots $-r_1, \dots, -r_C$ of B are simple, partial fractions give real constants

$$c, b_1, \dots, b_C \in \mathbb{R}$$

such that

$$R(z) = c + \sum_{j=1}^C \frac{b_j}{z + r_j}.$$

Substituting $z = t(x)$ gives

$$R(t(x)) = c + \sum_{j=1}^C \frac{b_j}{t(x) + r_j}.$$

This score has the correct signs on the cube:

$$f(x) = 1 \iff F(t(x)) = 1 \iff R(t(x)) > 0.$$

Step 3. Each reciprocal term is one head

It remains to show that, for any fixed $r > 0$ and $b \in \mathbb{R}$,

$$x \mapsto \frac{b}{t(x) + r}$$

is a valid one-head atom.

Let

$$\Lambda := \sum_{i=1}^n \lambda_i.$$

Choose

$$\alpha > 1 + \frac{\Lambda}{r}, \quad \rho_i := \frac{\lambda_i}{\alpha - 1}, \quad \gamma := r - \frac{\Lambda}{\alpha - 1}.$$

Then

$$\alpha > 1, \quad \rho_i > 0, \quad \gamma > 0.$$

On the Boolean cube,

$$\alpha^{x_i} = 1 + (\alpha - 1)x_i.$$

Therefore

$$\begin{aligned} \gamma + \sum_{i=1}^n \rho_i \alpha^{x_i} &= r - \frac{\Lambda}{\alpha - 1} + \sum_{i=1}^n \frac{\lambda_i}{\alpha - 1} (1 + (\alpha - 1)x_i) \\ &= r + \sum_{i=1}^n \lambda_i x_i \\ &= r + t(x). \end{aligned}$$

Now take the numerator of the one-head atom to be the constant b , by setting

$$\eta = b, \quad m_i = 0, \quad \delta = 0.$$

The atom becomes exactly

$$\phi(x) = \frac{b}{r + t(x)}.$$

Thus each summand

$$\frac{b_j}{t(x) + r_j}$$

costs one head. The constant c is absorbed into the final readout threshold. Consequently the score

$$R(t(x)) = c + \sum_{j=1}^C \frac{b_j}{t(x) + r_j}$$

is realized with C heads and computes f after thresholding at zero. Hence

$$H^*(f) \leq C_t(F).$$

Minimizing over all positive-parallel representations gives

$$H^*(f) \leq P_{\parallel}^+(f).$$

□

What the positivity assumption is doing

The proof used $\lambda_i > 0$ in exactly one place: it made the denominator

$$t(x) + r = r + \sum_i \lambda_i x_i$$

positive on the cube with all slopes of the same sign. That is precisely the kind of denominator produced by a single softmax attention head in the scalar normal form above.

If one allows arbitrary parallel hyperplanes

$$L(x) = a_0 + \sum_i a_i x_i$$

with mixed-sign coefficients a_i , the same partial-fraction proof would produce terms of the form

$$\frac{b}{L(x) + r}.$$

Here $b \in \mathbb{R}$ is a numerator coefficient and r is a shift chosen so that the denominator avoids zero on the cube. Those denominators may have mixed-sign slopes. They are not automatically one-head denominators. Thus the theorem proved here is the positive-parallel bound

$$H^*(f) \leq P_{\parallel}^+(f),$$

not an unrestricted mixed-sign parallel bound.